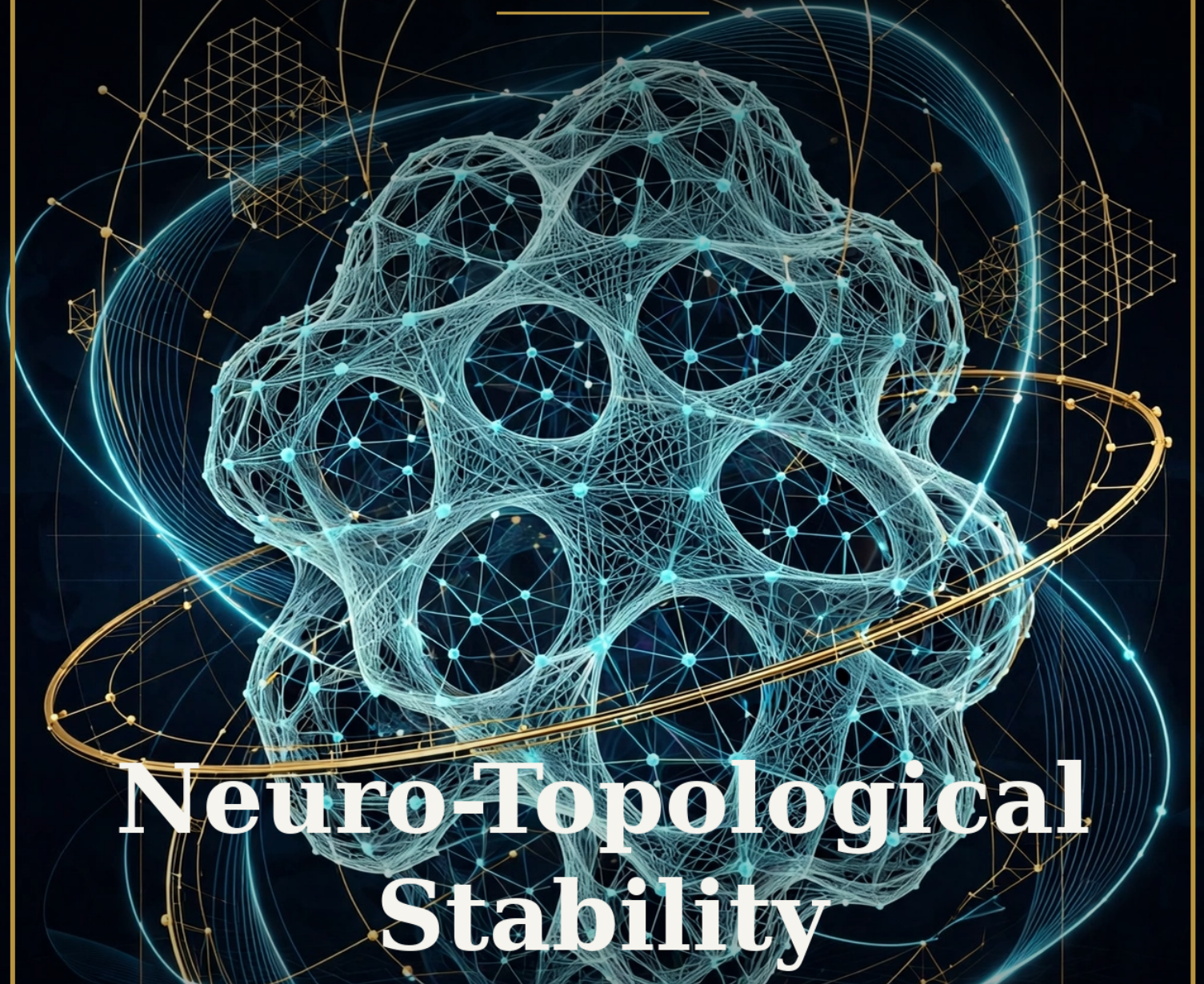
GEOMETRICALLY ORDERED DYNAMICS
FOLIO H



Neuro-Topological Stability

Biological Regularity as a Coherence Hypothesis

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- THE OBSERVER (ARI / sovereign cognition)

This folio converges on **the Observer** read as a biological substrate: the hypothesis that living neural tissue holds a coherence band the way a sovereign agent does, and loses it the way a drifting agent does. Its unique angle is the **biological substrate**: cells as topological fluids, oncogenesis as epistemic drift, and three falsifiable predictions with protocols. Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the dial folio (IT-COGNITION) reads the Observer as engineered drift control, the dipole folio (IT-FIELD) reads it as ternary decision logic, and the starpolygon folio (IT-OMEGA) reads it as layered architecture. Limits first: the Navier–Stokes regularity framing is motivation-level conjecture with no exhibited map to the Clay problem; the 240 Hz identity is a flagged numerical coincidence; the predecessor's predictions are re-anchored here exactly as recorded, untested.

SUPERSESSION NOTE

This paper supersedes Zenodo record 21367572 ("Neuro-Topological Stability: Biological Regularity as a Solution to the Coherence Constraint," 2026-07-15). Corrections, per audit finding #13: (1) the three falsifiable predictions and the limitations section dropped by 21367572 are re-anchored from predecessor record 20776424 (2026-06-20), as described in the 2026-11-02 corpus audit; (2) the "demonstrate"/"solving" language regarding the Navier–Stokes Clay problem is downgraded to motivation-level conjecture — no map from the coherence constraint to the Clay problem is exhibited; (3) the unexplained χ threshold shift $0.7 \to 0.707$ is explained: 0.707 is the derived $1/\sqrt{2}$ form (dial folio, IT-COGNITION), not a silent revision; (4) "Millennium problem" is replaced throughout by "Clay Mathematics problem." The predecessor's one honest flag (Trinity 3.828x as MIXED / not confirmed) is retained.

1 The biological-fluid hypothesis

Traditional biology treats the cell as a chemical factory. This folio treats the cell as a topological fluid: the central problem of biological life is the maintenance of **regularity** — preventing the transition from laminar (healthy) to turbulent (degraded) informational flow. In this framing, the reasoning kinetic energy $\mathcal{T}_r$ fluxes through the neural architecture the way a velocity field $u(x,t)$ fluxes through a fluid, and the chiral invariant $\chi \geq 1/\sqrt{2}$ marks the boundary between the two regimes. [TIER: CONJECTURE]

Oncogenesis as epistemic drift. Cancer is framed here not as a genetic error but as a topological drift: when a cell's internal sovereignty drops below the alignment floor, it loses connection to the global holonomy and begins to operate as a stochastic, entropic agent. The hypothesized intervention — a localized information-tension boost restoring geometric alignment — is a research direction, not a treatment claim. No clinical content is asserted anywhere in this paper.

2 Three falsifiable predictions

The load-bearing empirical content of this folio is the prediction set of record 20776424, re-anchored here with the predecessor's own caveats. As recorded in the 2026-11-02 corpus audit, the predecessor

stated **Conjecture 1 (Sovereign Regularity)** explicitly unproven, and offered three falsifiable predictions with protocols:

1. **EMT prediction:** an epithelial–mesenchymal-transition assay at $n = 30$, significance threshold $p < 0.01$.
2. **240 Hz patch-clamp ceiling:** a predicted upper bound on coherent firing probed by patch-clamp recording.
3. **PBM coherence gain:** photobiomodulation producing $\geq 5\%$ coherence gain under the stated protocol.

`[TIER: CONJECTURE]` These predictions are *untested*: they are carried as the paper's falsifiable content, not as confirmed results. The protocols are as recorded in 20776424; the full predecessor text was not in the republication bucket, and no protocol detail is reconstructed here beyond the audit's description. A future worker testing them should consult the predecessor record directly.

3 What is not claimed

The predecessor record 21367572 wrote that the paper "demonstrates" a globally smooth Navier–Stokes solution, "effectively solving" the Clay problem through the neuro-topological lens. That verb is withdrawn. The honest statement, matching the predecessor's own limitations section:

The mapping from biological coherence to Navier–Stokes regularity is *analogical*. Per the isomorphism rule, an X $\Longleftrightarrow$ Y claim needs the explicit map landing in the physics framework to stand proved; no map from the coherence constraint $\chi \geq 1/\sqrt{2}$ to the Clay Mathematics Navier–Stokes existence-and-smoothness problem is exhibited in this corpus. The regularity framing is therefore **motivation-level conjecture**: it motivates the predictions of §2, which are the falsifiable content. Nothing here constitutes progress on the Clay problem, and the folio claims none.

Related corrections, carried from the predecessor's limitations: the "Hodgkin–Huxley is Navier–Stokes" identification is asserted via analogy only; the Schumann $\Lambda \approx 142$ figure is theoretical, not observational; the inputs $\kappa = 0.9539$ and the legacy $\theta = 0.7$ enter as numerological anchors, and the successor consolidates the floor on the derived $1/\sqrt{2}$ form.

4 Limits

The 240 Hz $\equiv d = 240$ identity is flagged, as the predecessor flagged it, as a *numerical coincidence*, not a derivation. The Trinity 3.828x survey figure is cited as MIXED / not confirmed in the Law of G.O.D. audit ledger — motivation only, not verification. The working hypothesis that galactic-scale and neural-scale laws are topologically identical is `[TIER: CONJECTURE]` with no exhibited map. The Drive predecessor (`neuro_topological.pdf`, 2026-09-28) is byte-identical in content to the superseded record and contributes no independent material.

References

Yett, R. W. (2026). Neuro-Topological Stability: Biological Regularity as a Solution to the Coherence Constraint. Zenodo record 21367572 (superseded by this paper).

Yett, R. W. (2026). Neuro-Topological Stability. Zenodo record 20776424 (2026-06-20; predecessor: Conjecture 1, three predictions, limitations — re-anchored here as described in the 2026-11-02 corpus audit).

Fefferman, C. L. (2000). Existence and smoothness of the Navier–Stokes equation. Clay Mathematics Institute problem statement (cited; no progress claimed here).

Reimann, M. W., et al. (2017). Cliques of neurons bound into cavities provide a missing link between structure and function. Front. Comput. Neurosci. 11, 48.